

TABLE I
MAIN RESULTS: MACRO-F1 ON THE HELD-OUT TEST SET, MEAN OVER 3 SEEDS (42, 7, 2024).

Model	Macro-F1
Single-view SVM (baseline)	0.5534
Standard co-training (no WNM)	0.5767
PMVC-WNM (deployed)	0.5794
Best exhaustive candidate, View A (ridge_calibrated)	0.5652
Best exhaustive candidate, View B (ridge_calibrated)	0.5542
Full-supervision ceiling ($\sim 9,430$ labels)	0.6562

TABLE II
VIEW A / VIEW B CLASSIFIER COMPARISON (MEAN OVER 3 SEEDS).
LOWER BRIER SCORE INDICATES BETTER-CALIBRATED CONFIDENCE ESTIMATES.

View	Classifier	Macro-F1	Brier	Fit (s)
A	svm_platt	0.5624	0.1807	1.435
A	logreg	0.5613	0.1802	0.729
A	svm_calibrated	0.5596	0.1813	0.120
A	svm_softmax	0.5534	0.1960	0.018
B	logreg	0.5522	0.1845	1.314
B	svm_softmax	0.5488	0.2002	0.007
B	complement_nb	0.5340	0.1952	0.005
B	random_forest	0.5245	0.1907	1.492

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TABLE III
FEATURE N-GRAM RANGE COMPARISON AT THE DEPLOYED 5,000-FEATURE BUDGET (MEAN OVER 3 SEEDS).

View	Config	Macro-F1
A	char(2,3)	0.5611
A	char(3,4) [deployed]	0.5534
B	word(1,1)	0.5575
B	word(1,2) [deployed]	0.5522

TABLE IV
PSEUDO-LABEL GATE COMPARISON, CONFIDENCE THRESHOLD FIXED AT 0.55 (MEAN $\pm$ STD OVER 3 SEEDS).

Gate	Macro-F1	Pseudo-labels	Purity
agreement_and_confident (pmvc_wnm)	0.5794 ± 0.0094	7191	0.859
view_a_only (standard)	0.5767 ± 0.0101	5783	0.881
agreement_or_one_very_confident	0.5739 ± 0.0058	21827	0.757
agreement_no_threshold	0.5701 ± 0.0077	35100	0.709
or_union	0.5701 ± 0.0077	35100	0.673

TABLE V
CUMULATIVE SINGLE-VIEW (VIEW A) IMPROVEMENT LADDER, EACH ROW ADDS ONE CHANGE ON TOP OF THE PREVIOUS (MEAN OVER 3 SEEDS).

Configuration	Macro-F1
0. baseline (svm_softmax, char(3,4), 5k feat)	0.5534 ± 0.0120
1. max_features=50000	0.5729 ± 0.0036
2. + min_df=2	0.5733 ± 0.0016
3. + class_weight=balanced	0.5741 ± 0.0024
4. + C tuned (C=0.3)	0.5751 ± 0.0028
5. + word(1,2) UNION char(3,5) [50k total]	0.5830 ± 0.0019
6. + CalibratedClassifierCV(LinearSVC)	0.5849 ± 0.0032

TABLE VI
TOP 5 OF 21 CANDIDATES TESTED, VIEW A (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
ridge_calibrated	linear	0.5652 ± 0.0064	0.6688	0.08
logreg	linear	0.5613 ± 0.0089	0.6599	1.40
bagging_linsvc	tree_ensemble	0.5601 ± 0.0013	0.6631	0.08
linsvc_calibrated	linear	0.5596 ± 0.0065	0.6642	0.08
knn_k15	instance_based	0.5541 ± 0.0069	0.6199	0.00

TABLE VII
TOP 5 OF 21 CANDIDATES TESTED, VIEW B (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
ridge_calibrated	linear	0.5542 ± 0.0076	0.6446	0.06
logreg	linear	0.5522 ± 0.0076	0.6381	0.58
linsvc_calibrated	linear	0.5519 ± 0.0074	0.6420	0.05
linsvc_softmax	linear	0.5488 ± 0.0065	0.6359	0.01
bagging_linsvc	tree_ensemble	0.5471 ± 0.0065	0.6377	0.05

TABLE VIII
LABEL-BUDGET SENSITIVITY: TEAM’S DEPLOYED MODEL VS.
BEST-FOUND MODEL (MEAN OVER 3 SEEDS).

Labeled seed	Team A	Best A	Team B	Best B
300	0.5376	0.5466	0.5176	0.5126
600	0.5534	0.5652	0.5522	0.5542
900	0.5669	0.5813	0.5701	0.5729
1200	0.5908	0.6017	0.5830	0.5848

TABLE IX
SPELLING-NOISE ROBUSTNESS: MACRO-F1 UNDER INJECTED
PHONETIC-VARIANT CORRUPTION (MEAN OVER 3 SEEDS).

Noise rate	Team A	Best A	Team B	Best B
0%	0.5534	0.5652	0.5522	0.5542
20%	0.5486	0.5614	0.5516	0.5535
35%	0.5444	0.5554	0.5524	0.5548

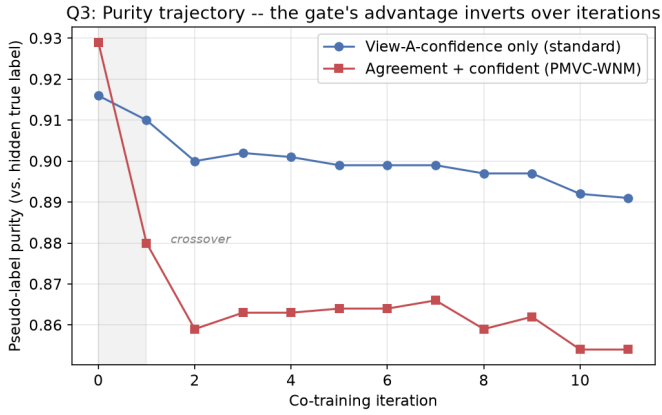


Fig. 1. Pseudo-label purity by co-training iteration, seed 42. The agreement gate’s purity advantage appears only at iteration 0 and inverts by the final iteration.

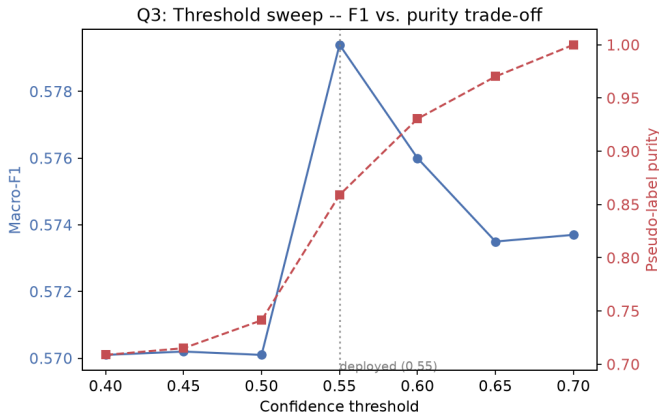


Fig. 2. Macro-F1 (left axis) and pseudo-label purity (right axis) vs. confidence threshold on the deployed agreement gate.

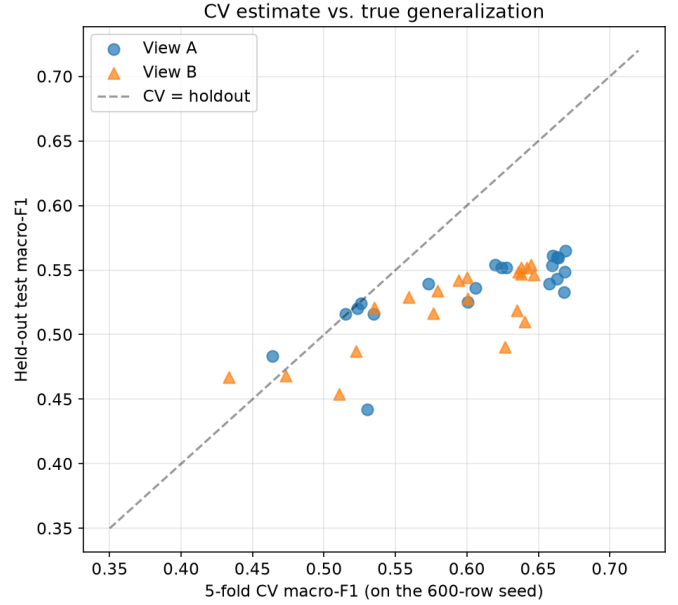


Fig. 3. Every candidate’s 5-fold CV estimate vs. its held-out test score. Points below the diagonal generalize worse than their CV estimate suggested.

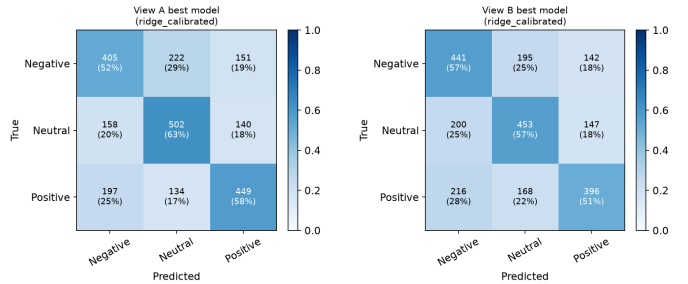


Fig. 4. Row-normalized confusion matrices, best model per view, seed 42.